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The Latest vs. the Greatest

Your stomach rumbles. Do you go to the Italian restaurant that you know and love, or the new Thai place that just opened up? Do you take your best friend, or reach out to a new acquaintance you'd like to get to know better? This is too hard—maybe you'll just stay home. Do you cook a recipe that you know is going to work, or scour the Internet for new inspiration? Never mind, how about you just order a pizza? Do you get your "usual," or ask about the specials? You're already exhausted before you get to the first bite. And the thought of putting on a record, watching a movie, or reading a book—*which one?*—no longer seems quite so relaxing.

Every day we are constantly forced to make decisions between options that differ in a very specific dimension: do we try new things or stick with our favorite ones? We intuitively understand that life is a balance between novelty and tradition, between the latest and the greatest, between taking risks and savoring what we know and love. But just as with the look-or-leap dilemma of the apartment hunt, the unanswered question is: what balance?

In the 1974 classic *Zen and the Art of Motorcycle Maintenance,* Robert Pirsig decries the conversational opener "What's new?"—arguing that the question, "if pursued exclusively, results only in an endless parade of trivia and fashion, the silt of tomorrow." He endorses an alternative as vastly superior: "What's best?"

But the reality is not so simple. Remembering that every "best" song and restaurant among your favorites began humbly as something merely "new" to you is a reminder that there may be yet-unknown bests still out

there—and thus that the new is indeed worthy of at least some of our attention.

Age-worn aphorisms acknowledge this tension but don't solve it. "Make new friends, but keep the old / Those are silver, these are gold," and "There is no life so rich and rare / But one more friend could enter there" are true enough; certainly their scansion is unimpeachable. But they fail to tell us anything useful about the *ratio* of, say, "silver" and "gold" that makes the best alloy of a life well lived.

Computer scientists have been working on finding this balance for more than fifty years. They even have a name for it: the explore/exploit tradeoff.

Explore/Exploit

In English, the words "explore" and "exploit" come loaded with completely opposite connotations. But to a computer scientist, these words have much more specific and neutral meanings. Simply put, exploration is *gathering* information, and exploitation is *using* the information you have to get a known good result.

It's fairly intuitive that never exploring is no way to live. But it's also worth mentioning that never exploiting can be every bit as bad. In the computer science definition, exploitation actually comes to characterize many of what we consider to be life's best moments. A family gathering together on the holidays is exploitation. So is a bookworm settling into a reading chair with a hot cup of coffee and a beloved favorite, or a band playing their greatest hits to a crowd of adoring fans, or a couple that has stood the test of time dancing to "their song."

What's more, exploration can be a curse.

Part of what's nice about music, for instance, is that there are constantly new things to listen to. Or, if you're a music journalist, part of what's terrible about music is that there are *constantly* new things to listen to. Being a music journalist means turning the exploration dial all the way to 11, where it's nothing but new things all the time. Music lovers might imagine working in music journalism to be paradise, but when you constantly have to explore the new you can never enjoy the fruits of your connoisseurship—a particular kind of hell. Few people know this

experience as deeply as Scott Plagenhoef, the former editor in chief of *Pitchfork*. "You try to find spaces when you're working to listen to something that you just want to listen to," he says of a critic's life. His desperate urges to stop wading through unheard tunes of dubious quality and just listen to what he loved were so strong that Plagenhoef would put only new music on his iPod, to make himself physically incapable of abandoning his duties in those moments when he just really, really, really wanted to listen to the Smiths. Journalists are martyrs, exploring so that others may exploit.

In computer science, the tension between exploration and exploitation takes its most concrete form in a scenario called the "multi-armed bandit problem." The odd name comes from the colloquial term for a casino slot machine, the "one-armed bandit." Imagine walking into a casino full of different slot machines, each one with its own odds of a payoff. The rub, of course, is that you aren't told those odds in advance: until you start playing, you won't have any idea which machines are the most lucrative ("loose," as slot-machine aficionados call it) and which ones are just money sinks.

Naturally, you're interested in maximizing your total winnings. And it's clear that this is going to involve some combination of pulling the arms on different machines to test them out (exploring), and favoring the most promising machines you've found (exploiting).

To get a sense for the problem's subtleties, imagine being faced with only two machines. One you've played a total of 15 times; 9 times it paid out, and 6 times it didn't. The other you've played only twice, and it once paid out and once did not. Which is more promising?

Simply dividing the wins by the total number of pulls will give you the machine's "expected value," and by this method the first machine clearly comes out ahead. Its 9–6 record makes for an expected value of 60%, whereas the second machine's 1–1 record yields an expected value of only 50%. But there's more to it than that. After all, just two pulls aren't really very many. So there's a sense in which we just don't yet *know* how good the second machine might actually be.

Choosing a restaurant or an album is, in effect, a matter of deciding which arm to pull in life's casino. But understanding the explore/exploit tradeoff isn't just a way to improve decisions about where to eat or what

to listen to. It also provides fundamental insights into how our goals should change as we age, and why the most rational course of action isn't always trying to choose the best. And it turns out to be at the heart of, among other things, web design and clinical trials—two topics that normally aren't mentioned in the same sentence.

People tend to treat decisions in isolation, to focus on finding each time the outcome with the highest expected value. But decisions are almost never isolated, and expected value isn't the end of the story. If you're thinking not just about the *next* decision, but about *all* the decisions you are going to make about the same options in the future, the explore/exploit tradeoff is crucial to the process. In this way, writes mathematician Peter Whittle, the bandit problem "embodies in essential form a conflict evident in all human action."

So which of those two arms should you pull? It's a trick question. It completely depends on something we haven't discussed yet: how long you plan to be in the casino.

Seize the Interval

"Carpe diem," urges Robin Williams in one of the most memorable scenes of the 1989 film *Dead Poets Society*. "Seize the day, boys. Make your lives extraordinary."

It's incredibly important advice. It's also somewhat self-contradictory. Seizing a day and seizing a lifetime are two entirely different endeavors. We have the expression "Eat, drink, and be merry, for tomorrow we die," but perhaps we should also have its inverse: "Start learning a new language or an instrument, and make small talk with a stranger, because life is long, and who knows what joy could blossom over many years' time." When balancing favorite experiences and new ones, nothing matters as much as the interval over which we plan to enjoy them.

"I'm more likely to try a new restaurant when I move to a city than when I'm leaving it," explains data scientist and blogger Chris Stucchio, a veteran of grappling with the explore/exploit tradeoff in both his work and his life. "I mostly go to restaurants I know and love now, because I know I'm going to be leaving New York fairly soon. Whereas a couple years ago I moved to Pune, India, and I just would eat friggin' everywhere

that didn't look like it was gonna kill me. And as I was leaving the city I went back to all my old favorites, rather than trying out new stuff.…Even if I find a slightly better place, I'm only going to go there once or twice, so why take the risk?"

A sobering property of trying new things is that the value of exploration, of finding a new favorite, can only go down over time, as the remaining opportunities to savor it dwindle. Discovering an enchanting café on your last night in town doesn't give you the opportunity to return.

The flip side is that the value of exploitation can only go *up* over time. The loveliest café that you know about today is, by definition, at least as lovely as the loveliest café you knew about last month. (And if you've found another favorite since then, it might just be more so.) So explore when you will have time to use the resulting knowledge, exploit when you're ready to cash in. The interval makes the strategy.

Interestingly, since the interval makes the strategy, then by observing the strategy we can also infer the interval. Take Hollywood, for instance: Among the ten highest-grossing movies of 1981, only two were sequels. In 1991, it was three. In 2001, it was five. And in 2011, *eight* of the top ten highest-grossing films were sequels. In fact, 2011 set a record for the greatest percentage of sequels among major studio releases. Then 2012 immediately broke that record; the next year would break it again. In December 2012, journalist Nick Allen looked ahead with palpable fatigue to the year to come:

> Audiences will be given a sixth helping of X-Men plus *Fast and Furious 6, Die Hard 5, Scary Movie 5* and *Paranormal Activity 5.* There will also be *Iron Man 3, The Hangover 3,* and second outings for *The Muppets, The Smurfs, GI Joe* and *Bad Santa.*

From a studio's perspective, a sequel is a movie with a guaranteed fan base: a cash cow, a sure thing, an exploit. And an overload of sure things signals a short-termist approach, as with Stucchio on his way out of town. The sequels are more likely than brand-new movies to be hits this year, but where will the beloved franchises of the future come from? Such a sequel deluge is not only lamentable (certainly critics think so); it's also somewhat poignant. By entering an almost purely exploit-focused phase, the film industry seems to be signaling a belief that it is near the end of its

interval.

A look into the economics of Hollywood confirms this hunch. Profits of the largest film studios declined by 40% between 2007 and 2011, and ticket sales have declined in seven of the past ten years. As the *Economist* puts it, "Squeezed between rising costs and falling revenues, the big studios have responded by trying to make more films they think will be hits: usually sequels, prequels, or anything featuring characters with name recognition." In other words, they're pulling the arms of the best machines they've got before the casino turns them out.

Win-Stay

Finding optimal algorithms that tell us exactly how to handle the multiarmed bandit problem has proven incredibly challenging. Indeed, as Peter Whittle recounts, during World War II efforts to solve the question "so sapped the energies and minds of Allied analysts…that the suggestion was made that the problem be dropped over Germany, as the ultimate instrument of intellectual sabotage."

The first steps toward a solution were taken in the years after the war, when Columbia mathematician Herbert Robbins showed that there's a simple strategy that, while not perfect, comes with some nice guarantees.

Robbins specifically considered the case where there are exactly two slot machines, and proposed a solution called the **Win-Stay, Lose-Shift** algorithm: choose an arm at random, and keep pulling it as long as it keeps paying off. If the arm doesn't pay off after a particular pull, then switch to the other one. Although this simple strategy is far from a complete solution, Robbins proved in 1952 that it performs reliably better than chance.

Following Robbins, a series of papers examined the "stay on a winner" principle further. Intuitively, if you were already willing to pull an arm, and it has just paid off, that should only increase your estimate of its value, and you should be only more willing to pull it again. And indeed, win-stay turns out to be an element of the optimal strategy for balancing exploration and exploitation under a wide range of conditions.

But lose-shift is another story. Changing arms each time one fails is a pretty rash move. Imagine going to a restaurant a hundred times, each

time having a wonderful meal. Would one disappointment be enough to induce you to give up on it? Good options shouldn't be penalized too strongly for being imperfect.

More significantly, Win-Stay, Lose-Shift doesn't have any notion of the interval over which you are optimizing. If your favorite restaurant disappointed you the last time you ate there, that algorithm always says you should go to another place—even if it's your last night in town.

Still, Robbins's initial work on the multi-armed bandit problem kicked off a substantial literature, and researchers made significant progress over the next few years. Richard Bellman, a mathematician at the RAND Corporation, found an exact solution to the problem for cases where we know in advance exactly how many options and opportunities we'll have in total. As with the full-information secretary problem, Bellman's trick was essentially to work backward, starting by imagining the final pull and considering which slot machine to choose given all the possible outcomes of the previous decisions. Having figured that out, you'd then turn to the second-to-last opportunity, then the previous one, and the one before that, all the way back to the start.

The answers that emerge from Bellman's method are ironclad, but with many options and a long casino visit it can require a dizzying—or impossible—amount of work. What's more, even if we are able to calculate all possible futures, we of course don't always know exactly how many opportunities (or even how many options) we'll have. For these reasons, the multi-armed bandit problem effectively stayed unsolved. In Whittle's words, "it quickly became a classic, and a byword for intransigence."

The Gittins Index

As so often happens in mathematics, though, the particular is the gateway to the universal. In the 1970s, the Unilever corporation asked a young mathematician named John Gittins to help them optimize some of their drug trials. Unexpectedly, what they got was the answer to a mathematical riddle that had gone unsolved for a generation.

Gittins, who is now a professor of statistics at Oxford, pondered the question posed by Unilever. Given several different chemical compounds,